Resilience Academy

SDI Training for ICT Professionals

Environment Setup

Installing Docker Desktop and QGIS — your SDI toolkit

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Day 1 | 30 Minutes | crd.resilienceacademy.ac.tz

Resilience Academy

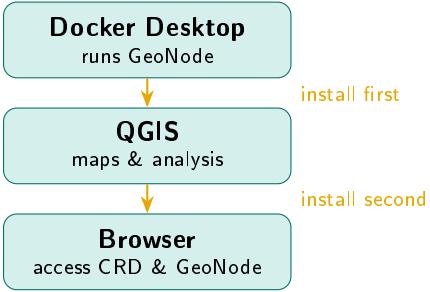
Session Overview — What We Install Today

This short lecture explains *what* and *why*.
The step-by-step installation instructions are in your **practical guide** (printed handout).

Three things to set up:

1. **Docker Desktop** — runs GeoNode on your laptop
2. **QGIS** — desktop application for viewing and analysing spatial data
3. **Web browser** — verify access to CRD

All three are **free**. All three have graphical installers — no command-line wizardry needed.



What Is Docker Desktop?

Docker Desktop is a **GUI application** (with an icon, a window, menus) that lets your computer run Docker containers.

What you see:

- An icon in your system tray (bottom-right on Windows, top-right on Mac)
- A dashboard window showing running containers
- Green status = Docker is running and ready

What it does for you:

- Handles all the Linux virtualisation automatically

Not Scary Terminal Stuff

Docker Desktop is installed like any other application:

1. Download the installer from **docker.com**
2. Double-click to install
3. Restart your computer
4. Open Docker Desktop from your applications menu
5. Wait for the green "Running" status

Windows Users

What Is QGIS?

QGIS is a free, open-source desktop application for working with maps and spatial data.

Think of it as: "Google Earth Pro on steroids" — but completely free.

What you can do with QGIS:

- **Open** Shapefiles, GeoTIFFs, GeoJSON, GeoPackage, CSV with coordinates — all the formats from Session 1
- **View** multiple layers on one map (e.g. roads + flood zones + schools)
- **Style** your map with colours, labels, and symbols
- **Connect** to CBD using WMS/WFS — live

Installation Is Simple

1. Go to **qgis.org/download**
2. Choose the **Long Term Release** (LTR) — the most stable version
3. Download for your OS (Windows / Mac / Linux)
4. Run the installer — accept defaults
5. Open QGIS from your applications

Which Version?

Always install the **LTR** (Long Term Release), currently QGIS 3.34 or newer

Installation Roadmap

Your practical guide has the detailed steps. Here is the big picture:

Step 1 Install Docker Desktop (20 min) → **Step 2** Restart & verify Docker (5 min) → **Step 3** Install QGIS LTR (15 min) → **Step 4** Open QGIS and verify (5 min)

If You Get Stuck

- Check the **practical guide** — it has screenshots for every step
- Ask your neighbour — they may have just solved the same problem
- Raise your hand — the trainers will come to help

Common Issues

- **"Virtualisation not enabled"** — need to change a BIOS setting (trainers will help)
- **Slow download** — use the offline installer from the USB drive provided
- **Mac permission popup** — click

Verification Checklist — Are You Ready?

After installation, confirm each item below. All three must pass.

1. Docker Desktop Works

- Open Docker Desktop
- Status bar shows **green** ("Docker is running")
- Open a terminal and type:

```
docker --version
```

You should see something like:
`Docker version 27.x.x, build abc123`

If you see "command not found", Docker did not install correctly.

3. Browser Accesses CRD

- Open Chrome or Firefox
- Navigate to: **crd.resilienceacademy.ac.tz**
- The CRD homepage loads
- You can see the dataset catalogue

All Three Green?

Docker running ☐
QGIS opens ☐
CRD loads ☐

Day 1 — What We Achieved

Session 1: Understanding Spatial Data & SDI

- Spatial data = data + location
- Vector (points, lines, polygons) vs. raster (pixel grids)
- SDI = the highway network for data sharing
- CRD: Tanzania's resilience SDI with 287+ datasets

Session 2: Technology Overview

- GeoNode = "WordPress for maps"
- Docker = portable software containers
- Your daily tools: QGIS, browser,

Session 3: Linux Basics

- 10 terminal recipes — copy-paste, do not memorise
- Navigated files and fetched CRD data from the terminal

Session 4: Environment Setup

- Docker Desktop installed and running
- QGIS installed and opens
- CRD accessible in the browser

Tomorrow — Day 2

GeoNode Deep Dive: We launch